

A History of
LATIN AMERICA
to 1825

Third Edition

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in collaboration with Jacqueline Holler

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[10] *ECONOMY: SHIPS AND SILVER*

We order and command the viceroys, presidents, governors, and ministers of Our royal treasury to take the greatest care in seeking the profit and increase of all that belongs to Us in the provinces under their administration, and to apply their entire attention and diligence to the exploitation and working of the mines, to the collection of Our royal imposts and to the remittance to these realms of what is gathered; proceeding with great diligence, not allowing withholdings or delays in any sum from one year to another ... The good administration and lawful growth of Our royal income (which will be most pleasant to Us) is fitting to the service of God, our Lord, and to the preservation of these realms. And we charge Our viceroys and presidents that, considering that this is the nerve and spirit that gives vigor and being to the Royal Estate, they gather with Auditors, Royal Officials, Ministers and other persons who may seem most fitting to achieve this end, and that they discuss and treat these matters, and the reduction of expenses in so far as possible, so that by this means, and others that they may arrive at, Our Royal income is increased, so that We may use it to attend to the needs of Our Monarchy ...¹

THIS exhortation from Philip III, directed in 1617 to his senior administrators in the Indies, suggests how crucial American income had become to the Spanish crown in the sixteenth century. The "service of God" was not just a rhetorical flourish, for American wealth, particularly in the form of surging silver production from the 1560s on, had been part of the inspiration for Philip II's commitment of Spain to the militant defense of Catholicism in Europe.² Indeed, the appearance of that wealth seemed providential, a sign that Spain should assume the role of temporal leader of the effort against the spreading Protestantism of northern Europe. By some measures American wealth was astonishing. Silver production from Potosí alone in 1592, worth about 5.6 million ducats, amounted to some 44 percent of the Spanish crown's average yearly spending on state business in Europe between 1593 and 1597.³ It is true, of course, that the crown did not receive all the silver produced at Potosí (or anywhere else) but only the royalty tax on it of about 21 percent. But the crown's complete revenue from America, including mining royalties, sales

taxes, Indian tribute payments, customs duties and many other levies, amounted, in years of high remittances from the Indies, to about a fifth of the total royal income during Philip II's reign (1556–98). And just as important as the absolute quantity of silver, or even more important, was the fact that a large shipment of bullion could reliably be expected every year – and indeed, until 1600, a shipment of increasing size. The predictability of the arrivals of silver made American “treasure” particularly valuable to Philip II as a source of financial leverage. With it as security he could raise loans to fund his political and military efforts in Europe.⁴

Though the full extent of America's wealth in silver was not apparent until the second half of the sixteenth century, the royal and individual hopes for bullion before then that grew from Columbus's initial reports on Hispaniola and Central America were not disappointed. The Greater Antilles, though the scene of much washing for gold in the first two decades of the century, do not seem to have yielded abundantly. But spoils from the mainland conquests were a different matter. Cortés's initial force in Tenochtitlan seized plentiful gold, but lost most of it in the lake during their flight from the city in mid 1520. Atahualpa's famous ransom in Cajamarca of the amount of gold and silver needed to fill a large room to the height of a raised arm was an even greater haul. Together with other booty it came to about 1.5 million pesos, the equivalent, perhaps, of 1.5 million ounces of silver. The king's fifth of this reached Spain.⁵ Other regions that proved rich in loot of worked gold were the Peruvian coast and highland Colombia.

In the early years after the conquests, the looting mentality expanded into one of extracting wealth through tribute collection. The ideal of most early *encomenderos* was to preside over lordly households and lands while being sustained by the tribute, in gold, food, cloth and other products, of the native communities granted to them. But not all *encomenderos* found their tribute receipts sufficient in quantity and variety, and far from all were content in a role of passive receipt of their Indians' products. Given the obviously widespread presence of gold (and, it was soon clear, silver also) in the rock formations of the mainland, it is no surprise that mining was one of the first enterprises to which they turned. The crown, too, was deeply interested in the mining of precious metals. Bullion was money. For the monarchy, indeed, even more than for colonists, American wealth always meant mining more than any other sort of production.

MINING

Mining in New Spain began within a decade of the conquest. Although some gold was extracted, chiefly in the south, it was soon clear that Mexico was richer in silver ores than in gold. The first silver strikes came c.1530, at Zumpango and Sultepec. Taxco and Tlalpujahua followed c.1534. All these were in central Mexico, within 150 kilometers or so of the capital. They produced encouragingly, especially as there was still no more ambitious standard

against which to set them. But the full extent of Mexico's silver wealth was not revealed until the mid 1540s, when Spanish exploration reached the first of great ore deposits of the northern plateau. This was at Zacatecas (1546), and from there other northern discoveries were made, at, for example, Santa Bárbara in 1567 and San Luis Potosí c.1592. At Guanajuato, on the border between central Mexico and the arid north, and well south of Zacatecas, silver ore seems to have been found c.1550. Guanajuato was overshadowed by other districts for almost two centuries, but finally rose to become the new Potosí of the empire, outstripping any other silver center in New Spain or South America by a wide margin after 1740.

The ores of Potosí itself, long the token of Spanish American wealth for Spaniards and other Europeans alike, came to light in 1545. The Spanish, with the Pizarro brothers in the forefront, had worked the nearby mines of Porco, itself a major pre-conquest source of silver, from 1538. The discovery of the Rich Hill of Potosí was an offshoot of that effort; and the scion very soon outgrew its parent. In the century after 1550, the Potosí district, with the Rich Hill always its main source of ore, produced about half of Spanish America's silver.

The Andes proved richer in gold than New Spain, with major mining centers developing at Carabaya (1542) in south-eastern Peru, Chachapoyas and Valdivia (c.1550) in northern Peru and central Chile respectively, and Zaruma (c.1560) in southern Quito. But it was New Granada that dominated gold production in the Andes, and everywhere in the empire, in the sixteenth century and later. Production began at Popayán, in the south, before 1540, and at Antioquia, in the north, in 1546. New Granadan gold output seems to have passed through a deep slump late in the seventeenth century, while still exceeding that of any other region, and then climbed strongly throughout the eighteenth.

At the beginning, Spaniards drew on native mining and refining techniques wherever they existed. In New Spain native knowledge of metallurgy was slight, except among the Tarascans in the center-west, who were skilled producers and workers of copper. From them early Spanish silver miners may have learned something. In the Andes the Spanish certainly took much from a long and developed native tradition of underground mining, smelting, and elaboration of refined metals. The Andeans were familiar with gold, silver, copper, and tin; from the latter two, they alloyed bronze. Silver was the Spaniards' main interest. Such was the native skill in producing it in the Andes that, until almost 1570, Spanish mine owners in effect rented their mines to Indian managers, who in turn hired workers to cut and refine the ore. Spanish iron and steel tools probably were an advance over native implements in work below ground. But the refining of silver in the central Andes, predominantly at Porco and Potosí, was done in long-standing native fashion using small furnaces of stone or clay named *guayras*. The Quechua word means "air," and the furnaces were so named because the fire inside them was fanned simply by the mountain winds rather than by bellows. The fuel used was either *ichu*, the rough grass of the highlands, or llama dung. This simple smelting method was cheap and efficient enough, when applied to high grade ores, to yield a profit for both native workers and Spanish mine owners.

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In New Spain, silver was also produced by smelting for some years after mines were first opened. The technology here, though, was European, consisting essentially of a small furnace in the form of a hollow, square column of stone, 1.5 to 2 meters high. The charcoal fire inside was blown by bellows. The whole apparatus was cheap and simple. Even where powered bellows were used, with the energy coming from a waterwheel or animals, the necessary machinery could be built of wood on patterns familiar from European mills for fulling wool or grinding grain. Some early Spanish colonists, at least, would have known of these designs, even if, as seems likely, there were few experienced miners among them. Almost certainly there was as well some early flow of mining and refining knowledge to New Spain from Germany, the European region most advanced in metallurgy in the sixteenth century. Miners from Silesia crossed the Atlantic to Santo Domingo in 1528 as part of the Welsers' colonizing effort in Venezuela. But they never reached Venezuela. Some went back to Europe, but fifty or so seem to have remained in the New World, and quite possibly moved on to Mexico. In the mid 1530s the factors in Seville of the other great German banking house of the day, the Fuggers of Augsburg, definitely sent German smelters to New Spain to provide help with difficulties in smelting silver.

Germans possibly played a part in bringing about a radical advance beyond smelting that was achieved in New Spain in the mid 1550s. This was the introduction of amalgamation of silver ores, a refining process that revolutionized silver production in Spanish America and that may rank, in its world-wide effects, as high as any other technical innovation made in the Americas since Europeans first went there. The principle of amalgamation – the curious capacity of mercury to bind to certain other metals, even when they are in the form of ores – had been known in the Old World at least since Roman times. It had been long used on a small scale, as, for example, in the recovery of gold and silver filings from the sweepings from a silversmith's floor. It probably passed early to America, since mercury has been found in the remains of the first Spanish town on Hispaniola, at Isabela, which suggests that gold may have been extracted there from alluvial sand or crushed rock by amalgamation. But it was in New Spain that mercury was first used to draw silver from its ores on what could be called an industrial scale. A Spanish cloth merchant from Seville named Bartolomé de Medina is given the main credit for the invention. In 1555 the viceroy granted him monopoly rights in the process in Mexico for ten years. Clearly associated with Medina, however, perhaps particularly in developing the machinery for crushing ore to the fineness that effective amalgamation required, was a German named Gaspar Lohmann. Both were active in Sultepec and the newly opened deposits at Pachuca in the mid 1550s.

Large-scale amalgamation appeared in New Spain not simply as the outcome of Medina's and Lohmann's inventiveness, but clearly as a response to changing local conditions. By the 1550s ores rich enough to be smelted at a profit were becoming scarce. They had mainly been surface deposits enriched by weathering, and were soon skimmed off. Enormous quantities of ore remained, obviously enough; but they were of lower grade, and furthermore

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had to be brought up from increasing depths. The ores' yield, therefore, dropped, and at the same time their cost rose. Medina and Lohmann's amalgamation process, with its capacity to refine great volumes of poor ores quite cheaply, was the answer. How eagerly the process was seized on is shown by that fact that over 120 refiners took it up before 1555 was out.⁶ In the following years it spread to nearly all Mexican silver centers.

For lack of records, amalgamation's immediate effect on Mexican production cannot be seen. But in Potosí it had demonstrably dramatic results. The same problems of declining ore yields and rising costs as had earlier afflicted Mexico were depressing output there by the late 1560s. Viceroy Toledo arrived from Spain in 1569 with orders to install the mercury process in Potosí. He indeed played an active part in its adoption there, but the miners of Potosí would doubtless soon have used it in any case. The first mills for refining by amalgamation were built in the town in 1572. That was also the year of lowest production since records began in 1549. But, with the help of the amalgamation process, output grew between 1572 and 1582 6.7 times (from 26,000 to 174,000 kilograms). By 1592, the peak year in Potosí's history, it had risen to almost 202,000 kilograms. Amalgamation, it is true, was not solely responsible for this boom. In the 1570s Toledo had also increased the labor supply, reorganizing the forced drafting of native workers to miners and refiners in Potosí. But without amalgamation those reforms would have served little. The extra workers might have mined much ore at low cost. But it could not have been refined profitably with either Andean or European smelting techniques.

Amalgamation processing was itself far from cheap to set up; but the large initial investment required paid off quickly. The fundamental need was for a means of crushing the ore finely, so that the mercury, when added, could come into close contact with as much of the mineral as possible. This was done everywhere with a mill consisting of four to twelve vertical, iron-shod stamps, which were lifted in turn by cams rotating on a heavy shaft. The motive power was a waterwheel, where sufficient flow and head were available, or teams of mules in dry areas, such as northern Mexico. Once milled, the ore was mixed with water and mercury, and sometimes other reagents, such as salt, iron filings, and copper sulfate, that empirical testing proved useful. In the Andes, the mixing was done in large tanks holding some 2,300 kilograms of ore. Sometimes these were heated from below to speed the reaction. In New Spain, wooden troughs served the same purpose in the sixteenth century. After then the common method was to spread the blended ore out on a paved court, a practice that led to the use of the term *beneficio de patio*, or "patio processing," for amalgamation in New Spain. After some weeks (the actual time depending on the ambient temperature and other, lesser, variables) the refiner would decide, from experience, that the combining of mercury with silver was complete. The entire mixture was then washed, usually in wooden vats, sometimes fitted with powered rotating paddles. The mercury-silver amalgam sank to the bottom, whence it was gathered up, and excess mercury physically squeezed from it. Heating the amalgam then made the remaining mercury evaporate off; a honeycomb of pure silver remained. Once amalgamation became commonplace,

refiners tried with some success to minimize the loss of mercury in the process. They found it possible to recover some of it by carrying out the heating of the amalgam beneath a metal or clay hood. The mercury vapor condensed on the inside of the hood into the metal's familiar liquid form, which could then be collected. The practice also had the advantage of reducing somewhat the risk of workers' being poisoned by mercury vapor, which is highly toxic.

Amalgamation was a long, many-staged process, and in its complexity a far cry from earlier refining by smelting. The amalgamation plant, known in the Andes as an *ingenio* and in New Spain as an *hacienda de minas*, was a large and expensive affair, with its crushing mill and associate driving mechanism, tanks or patio, washing vats, store rooms, living quarters for workers and supervisor, and often enough a small chapel. The water-powered *ingenios* built in Potosí in the 1570s cost on average 50,000 pesos each, the price of 50 modest houses, or 5,000 llamas. Such installations were, along with mine workings themselves, and ocean-going trading vessels, the largest capital investments made by colonists in Spanish America.

This need for heavy investment meant that refining silver by amalgamation was a business pursued almost exclusively by the white portion of the colonial population. In New Spain, where from the beginning native people had done little more in mining than provide simple labor, the arrival of the new technique brought little change for Indians. But in the central Andes it meant an end to the control of extraction and refining of ores that many skilled native miners had enjoyed, along with some share of the profit, from the 1540s to the late 1560s. These men did not have access to the cash needed to build *ingenios*. Further, the technology of waterwheels, geared machinery, and iron and steel was an import, and alien to them. Indians in Potosí and elsewhere, then, did not become *azogueros* ("mercury men"), the owners of refining mills and the dominant force in silver production after the 1570s. Production of silver by Indians, nevertheless, did not stop at that point. *Guayras* long continued to burn on Andean mountainsides, smelting small pieces of rich ore that indigenous men, and women, picked out from mine tailings, or that underground workers took from the mines as part of their wages. In aggregate this smelted production may have been considerable, though it is impossible to estimate its amount. The same happened in Mexico, where Indians and *mestizos* were always to be found operating small smelting furnaces in and around Spanish-dominated mining towns.

By Spanish law, subsoil rights in America remained with the crown. Private possession of land included only the surface. But individuals might claim mines and then work them for their own profit so long as they paid a royalty to the treasury. The basic rate of the tax in the sixteenth century was a fifth of the metal produced, the *quinto real*. Reductions in the tax were granted from mid century onwards to stimulate mining. New Spain was seen as particularly needing this help, so that in the second half of the century (and later) the prevailing rate there was a tenth. It is mainly from these royalty records, which become more complete as the century progresses, that mining production has been calculated. The best such estimate to date places Spanish American silver

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Gold production the most complete estimate in silver pesos, for New Of the 33.11 million million, or 67.5 percent

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Table 10.1 Gold production in New Granada, New Spain, and Peru, 1550-1620

<i>Period</i>	<i>Amount</i> <i>(silver pesos, millions)</i>
1561-70	2.75
1571-80	2.78
1581-90	2.28
1591-1600	4.77
1601-10	6.88
1611-20	6.98
Total	33.11

output from its start until 1600 or 1610 at 375-400 million pesos, or 10.6-11.3 million kilograms.⁷ By far the greater part of this amount was produced after 1550, as Potosí and the north Mexican mines began to yield. The rate of increase of total Spanish American production in the period from 1560 to 1620-30 was the highest of any long period in the colonial era, at about 2.3 percent annually. Potosí, once amalgamation was adopted there, was mainly responsible for the high volume of output, and for a time for its growth also. Production at Potosí leveled off, however, in the 1590s and then began to fall in the new century, as ore quality declined and the costs of extraction rose with the increasing depth of the mines. But it was still the largest silver producer in the Indies, and its output did not subside to 1572 levels until the early eighteenth century. Production in New Spain continued to rise until the 1620s, before dropping slightly for perhaps forty years. There, several substantial mining districts were always in operation, and a decline in one or two of them did not much affect the total output.

Gold production grew also in the 1550-1620 period, but less quickly. The most complete estimate to date shows totals, converted to an equivalent value in silver pesos, for New Granada, New Spain, and Peru combined (table 10.1). Of the 33.11 million total for the whole period, New Granada produced 22.36 million, or 67.5 percent.⁸

Not only did gold output grow less quickly than silver's in the second half of the sixteenth century, but its value was far less. That remains true even if the earlier gold booty of the conquests is taken into account. The total amount of gold gathered by the Spanish in New Granada, New Spain, and Peru from 1521 to 1610 has been estimated at the equivalent of 48.46 million silver pesos.⁹ That is only 12 percent of the proposed c.400 million peso silver production of Spanish America to 1610. It was probably at some point in the 1540s, when little gold was left to loot, that the value of silver mined passed that of gold acquired. The lure of gold may first have drawn Spain and Spaniards to America; but it was the reality of silver that kept them there.

The third economically important metal mined in the colonies was mercury, obviously a strategic material once amalgamation became the normal method of refining silver ores. Constant searches for mercury deposits in New Spain

revealed no useful source there. But Peru was a different matter. From Huancavelica, in the mountains 240 kilometers south-east of Lima, native people had long taken cinnabar (red mercuric sulfide) for cosmetic uses, and the Spanish subsequently found that it was a deposit large enough to supply the mercury demands of Potosí and other Andean mines for most of the colonial era. Its output was not, though, sufficient to cover the Mexico's needs as well. Fortunately, Spain itself held one of the world's largest mercury sources, at Almadén, north of Córdoba. Mining there went back to Roman times, but much ore remained to be extracted. For most of the span of the empire, Almadén supplied Mexican silver producers with the mercury they needed. As the sources of a material of vital interest to Spain, the mines of both Almadén and Huancavelica were placed under state administration, although contractors from whom the crown bought mercury at negotiated prices actually worked them for many years (Almadén until 1645, and Huancavelica until 1782). The mercury produced was distributed through the royal treasury system, and sold at prices fixed by the government. This royal monopoly of mercury was created partly to ensure supply, partly to produce a profit, and partly to force silver miners to pay the royalty due on silver – since, in law, a treasury office would sell them no more mercury until they had paid the fifth or tenth they owed on silver previously refined. In practice this system was less than wholly effective. Some silver escaped taxation, though nobody knew, or knows, how much; and some mercury escaped from the monopoly into private trading channels.

By the end of the sixteenth century, silver-mining was clearly the most heavily capitalized productive activity in Spanish America, the one showing most pronounced specialization and division of labor, and the one in which imported technology played the largest part. (Amalgamation, though developed industrially in New Spain, is taken here as an imported technology, since it had no native American roots.) The large mining towns would be instantly recognizable to a time-traveler from the present as industrial centers: smoke, smells, polluted water, noise, bustle, contrasting opulence and poverty, sheer size. By 1600 Potosí had some 80 thudding refining mills in operation, spread along several kilometers of the small river flowing through the town. To prolong milling beyond the rainy summer months, a series of interconnected dams had by then been built in a massif to the east of the town: another large investment. In New Spain, Pachuca had 49 operating mills c.1597; Taxco, 47; Zacatecas, 20. The entire colony had 372 active refineries – *haciendas de minas* – in which were installed 399 animal-driven stamp mills and 205 powered by water. Almost all the *haciendas* with waterwheels were in the center and west, where substantial rain falls, at least in the summer.¹⁰

CLOTH

Among other productive activities in the sixteenth-century Indies, those closest to mining in their degree of technical innovation and their reliance on the formation of fixed capital were cloth- and sugar-making. In textiles, there was a

native tradition to draw on almost everywhere, and some regions, such as central Mexico and coastal Peru, had for many centuries past produced cloth of outstanding fineness and design. The Mexicans spun and wove cotton, lacking any wool-bearing animal. The Peruvians also used cotton; but their most remarkable weaving was of llama, alpaca, and vicuña wool, which they first spun to an astonishing fineness, a quality all the more remarkable in that it was achieved on simple, hand-held spindles. For weaving there was the backstrap loom, and an even simpler arrangement in which the warp is simply stretched between pegs set in the ground.

These looms, and the simple spindle, remained in use after the conquest, and indeed can still be seen in the Andes; men driving llama trains, for instance, spin as they walk along, twirling the spindle with one hand and feeding the wool onto it with the other. But the Spanish saw prospects of profit in quicker production, and soon imported European apparatus for that purpose, notably spinning wheels and treadle looms. These were devices of thirteenth-century origin, and in Spanish America continued in use until the nineteenth, finally succumbing to the power-driven machinery typical of the Industrial Revolution. But in sixteenth-century America they were radically new, and led to greater division of labor and much increased output.¹¹ They first came to Mexico in the 1530s, and followed on the conquerors' heels to South America. Complementing this technical advance, and accentuating its alien quality, was the arrival of two new fibers, of which sheep's wool was the more important in the long term. Its availability was a radical departure everywhere except in the central Andean natural range of the American camelids. Viceroy Mendoza himself brought the renowned Spanish merino sheep to New Spain, and ran flocks for his own profit. The more coarsely fleeced *churro* sheep had arrived sooner, via the Caribbean islands, where it had been introduced early, though without prospering.¹² Indians in both Mexico and South America quickly took to raising the animals themselves. Mendoza also made efforts to create a silk-raising and -weaving industry in New Spain, drawing on the great silk-working tradition of Moorish Spain. In this effort, Cortés was another keen participant. Silkworms were imported and mulberry plantations created to feed them. For much of the sixteenth century the production of silk thread prospered, especially in the south, and weaving thrived in Mexico City and Puebla. Finally, though, with the rise of the trans-Pacific trade in the 1570s, competition from oriental silks overcame the Mexican effort.¹³

The production of thread and cloth with the new Spanish machines tended early to concentrate in workshops, commonly termed *obrajes*. These were generally premises in which carding, spinning, and weaving were carried out under a single roof. *Obrajes* are mentioned at Puebla as early as 1539, less than a decade after the town was founded. The inventory of Cortés's estate, drawn up in 1549, shows that he had built one near Cuernavaca, his seat from the early 1530s. This establishment contained a water-powered fulling mill, four functioning looms, and 21 spinning wheels. The labor force of carders, spinners, weavers, and fullers is estimated at 40 to 50. They worked both merino and coarser wool.¹⁴ Fifty years or so later, mechanized textile production was a

salient feature of the Mexican economy. In 1604, between 98 and 130 *obrajes* and smaller workshops called *trapiches* were producing cloth. They were widely distributed across the center and west of New Spain, with large clusters in and around Puebla, Mexico City, Texcoco, and Tlaxcala. The workforce in each is thought to have averaged about 50, so that the total number making cloth by these imported methods was perhaps 6,000. This was two-thirds of the number of workers in Mexican silver-mining in the same years.¹⁵ How much cloth was produced, both in absolute quantity and in relation to native production by traditional methods, it is impossible to say; but clearly spinning and weaving in *obrajes* had become, by the late sixteenth century, a major component of the European side of the colonial economy. It had grown with backing from the crown, which was keen to ensure an adequate supply of affordable cloth in the colony.¹⁶ Late in the century qualms arose about competition with producers in Spain, but not to the point where effective restrictions were applied to Mexican production.

Obrajes appeared in Peru soon after the conquest and multiplied there, again with royal support, in the mid 1500s. The emigration of Spanish master weavers and accompanying carders, fullers, and so on was encouraged. Most major towns soon had their *obrajes*, where cloth continued to be made for the rest of the colonial period.¹⁷ Still, it was not Peru but Quito that became the leading textile region of South America. The province's central highlands offered great expanses of good grazing for sheep. And once local gold production began to decline, from about 1560, cloth-making seemed to leading settlers (most of them *encomenderos* still, in this rather isolated region) a promising means of acquiring cash. Fabric was a product compact and valuable enough to bear the high freight cost of export from Quito. Quito, indeed, for the rest of the colonial period depended on external textile sales to earn the money it needed to buy goods from elsewhere, and particularly from Europe; without cash it had no entry into the transatlantic trade. Its textile makers found they could sell cloth northwards to New Granada, for gold, and southwards to Peru and Charcas, for silver. The opulent age of *quiteño* cloth production was the seventeenth century. But the first *obrajes* probably date from the 1560s, when *encomenderos* and *caciques* joined forces to set up the workshops in native villages. The Spaniards looked for cash gains from sales. The native leaders, in return for organizing labor, took a salary from the *obraje*. But equally attractive to them was the fact that some of the profit went to pay their subjects' tributes, for which they were personally responsible. Thus native "community *obrajes*" came into being in Quito. By the opening of the seventeenth century fourteen of them existed in the highlands. Some of them had been founded, at least, on a notably large scale, with 200 or more workers. They remained one major base of textile production in Quito until the early eighteenth century. The other base consisted of shops owned privately, as in Mexico or Peru, that landowners set up on large rural estates where the primary raw material, sheep's wool, was produced. Between 1601 and 1628 the crown granted thirty-eight permits for the building of such *obrajes* in Quito.¹⁸

SUGAR

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SUGAR

If cloth manufacture was often sited in cool highland areas suited to running sheep, the warmth needed for growing cane meant that sugar production was generally a lowland activity. The Atlantic islands settled by Portugal and Spain in the 1400s had generally proven welcoming to cane, and sale of sugar to the mainland had become good business, as Columbus learned in visits to Madeira before 1480. Planting cane, then, was logically among the first ventures in commercial farming tried in Hispaniola. The first mill was built there, at Concepción, in 1503. It was only after 1515, however, that, with the yield of gold placers declining, and the sugar price in Europe rising, settlers on Hispaniola gave serious attention to sugar. A strip of coast immediately west of Santo Domingo was planted with cane. The first sugar refined emerged from a small, animal-powered mill, or *trapiche*. In 1517 the first water-driven mill (*ingenio*) came into action. The design of its essential component, the cane-crushing rollers, had its origins in Sicily. It had been used in the Canaries, from where men were now brought to build and run mills in Hispaniola. Two vertical wooden rollers, geared together in counter-rotation, were the crux of the design. The cane was drawn between them, and its juice squeezed out as it went.¹⁹ This device passed to the mainland, although there a three-roller pattern seems to have been used from the beginning, with the central roller driving, through gears mounted above, two others on each side. This arrangement gave the outer rollers opposing rotations, so that a piece of cane could be passed through in one direction, and then immediately back in the other to squeeze more juice from it.

Hispaniola thus became the first American colony exporting (to Spain) a plantation product. The other large islands followed suit in due course. Sugar production in New Spain, conversely, was largely for local consumption. Cane and milling technology had been carried there by 1530 from the Caribbean. The lower valleys around Cuernavaca and the coastal lowlands near Veracruz were the first good territory found for the crop. Then, as the century progressed, sugar planting extended westward into Michoacán, and Colima on the Pacific coast. Cortés, again, is the most famous of the early Mexican sugar raisers. His first mill (c.1529) at Tuxtla, on the Gulf coast, was probably the first in the colony. He built another, with an accompanying plantation, near Cuernavaca in the 1530s. The area around Cuernavaca became the major producing region, from a combination of climate, fertility, and above all closeness to the great market provided by Mexico City. The sweet tooth that New Spain – and the rest of Spanish America – developed in the sixteenth century constantly amazed newcomers and visitors from Europe, though it was a time in which the demand for sugar in Europe itself was apparently insatiable, as the largest American exporters of all, the planters in Brazil, found to their benefit. Indians as well as settlers took eagerly to sugar in New Spain, the natives probably consuming the less refined forms and the by-products such as molasses, while the clayed, whiter end-product went to colonists who could afford its higher price. By the early seventeenth century, fifty or sixty large *ingenios* and

trapiches were at work in Mexico, producing perhaps 3.5–5 million kilograms a year; and many smaller mills besides.²⁰

Some early Mexican sugar also went to Peru, though this trade had subsided by 1560 as a result of rising local output and a consequent drop in price. The first small sugar *trapiches* in Peru date from c. 1540. A reliable source, Pedro de la Gasca, states that four were running in 1549. From then on production rose to meet a local demand as impressive as Mexico's. (In 1542 the city council of Lima had banned the making of candied fruit because it caused harm to the republic and turned men into idle vagabonds. Underlying, perhaps, what seems a curious puritanism were the facts that sugar was dear, it was valued as a medicine, and that importing it from Mexico drained scarce coin from Peru.)²¹ From the start, Peruvian cane-growing was concentrated in the irrigated river valleys crossing the north and central coast. The great age of the plantations that grew up in those green patches amidst the desert was the seventeenth century, when the Peruvian mills supplied sugar to much of the west coast of South America and the central Andean highlands also.

The sugar refinery was a plant as complex and expensive as a silver-refining mill. The roller mechanism for expressing juice from the cane was a simpler device than the stamp mill; but the drive trains needed to carry energy from the waterwheel or animal team to the crushing machinery were similar in both cases. In place of the patio or refining tanks the sugar mill had a boiling house, equipped with large copper vats, and a purging house, with possibly hundreds of molds, where the boiled-down juice was formed into loaves. (The process is described in some detail in chapter 13, below, on colonial Brazil.) The apparatus, the buildings needed to hold it, and attached store rooms and living quarters were a costly proposition. In sixteenth-century Mexico a sugar mill cost 50,000 pesos; in the mid seventeenth century, refineries owned there by the regular clergy sold for up to twice that sum.²²

DYES

Sixteenth-century sugar-making in New Spain has been described as Mexico's "first agro-industry,"²³ and the same may be said of it for Peru. It was not the only such industry, though certainly the largest. From the early 1560s New Spain also produced indigo, a rich blue dye extracted from leguminous plants of the genus *Indigofera*. Production began near Cuernavaca, but soon spread southwards into warm southern areas and Yucatan. To extract the dye, the leaves had to be cooked and mashed. Preparing indigo on a large scale thus required boilers and crushing machinery, sometimes driven by a waterwheel. The mills, of which more than 48 existed in Yucatan by the late 1570s, were therefore substantial pieces of fixed capital, though an order of magnitude smaller than sugar refineries in cost. Mexico sent almost 7,000 kilograms of indigo to Spain in 1576. In 1609 the export was 132,000 kilograms, worth almost 550,000 pesos, a very large sum indeed.²⁴

Another dye was a still more valuable export of New Spain from the early years. This was cochineal, a red powder made by crushing the dried body of a

small insect living however, from indigo indeed any import throughout the colony pulverizing the insect in the producing region of the Atlantic. The first reached Antwerp in perhaps 113,000–160,000 pesos.²⁵ The second export in volume

SHIPBUILDING

A very different product embodying, in part, Cortés organized similarly Tehuantepec and found the necessary surprising. But the samples of the genre. In 1521 Cortés had López, to build the siege of Tenochtitlan that developed as the most of the sixteenth at Balboa's order, for that Pizarro used for building rose c. 153 Nicaragua to Panama them or more slave the prime reason for wood for hulls and particularly of wood southern Quito, as a major producer of slaves to the movement of tury, Guayaquil and vessels for the cons with the settlement

CRAFTS

Ocean-going ships technical innovation

small insect living on the nopal cactus. The preparation of cochineal differed, however, from indigo extraction in that it used no mechanical processing, or indeed any imported technology at all. Pre-conquest practices persisted throughout the colonial period, with Indians tending, collecting, drying, and pulverizing the insects. Southern Mexico, particularly the Oaxaca region, was the producing region. Spaniards bought the finished dye, mainly for sale across the Atlantic. The first export to Spain was in 1526; the first Mexican cochineal reached Antwerp in 1552, and London by 1569. By 1600 or so, Spain received perhaps 113,000–136,000 kilograms of Mexican cochineal yearly, worth some 600,000 pesos.²⁵ Throughout colonial times the dye remained New Spain's second export in value, behind (though far behind) silver.

SHIPBUILDING

A very different product of southern Mexico, and one relying on, indeed embodying, imported technology, was ships. From 1522 to the late 1530s, Cortés organized shipbuilding at different places on the Pacific coast – primarily Tehuantepec and Acapulco – for projected explorations. That he could have found the necessary carpentry skills among his men so early might seem surprising. But that simply indicates how far bands of conquistadores were samples of the general Spanish population rather than professional soldiers. In 1521 Cortés had made excellent use of one carpenter-conqueror, Martín López, to build the brigantines that gave him control of the lake during the final siege of Tenochtitlan. It was not so much Mexico, however, as Central America that developed as the main shipbuilding area on the American west coast for most of the sixteenth century. The first vessels were made in Panama in 1517, at Balboa's order, for exploration. Others followed in the 1520s, including those that Pizarro used for southward exploration towards Peru. Then the pace of building rose c.1530 to supply the needs of a new trade in Indian slaves from Nicaragua to Panama. By 1533 over thirty ships were sailing the Pacific, half of them or more slavers. Nicaragua's being the source of the slaves was certainly the prime reason for shipbuilding's rise there; but the region also had abundant wood for hulls and masts, fibers for ropes, and pitch.²⁶ Wealth of raw material, particularly of wood, later underlay also the development of Guayaquil, in southern Quito, as a shipbuilding center. By the late sixteenth century it was a major producer of shipping. Since passage of Cape Horn was a fearful obstacle to the movement of Atlantic shipping into the Pacific until the eighteenth century, Guayaquil and Central America were long almost the sole suppliers of vessels for the considerable trade that developed up and down the west coast with the settlement of Quito, Peru, and Chile.

CRAFTS

Ocean-going ships were the among the most striking examples of European technical innovation in sixteenth-century America, far exceeding the canoes of

the native Caribbean or the large rafts of pre-conquest north-west South America in their ability to move cargoes and use the sea as a relatively rapid, multi-directional highway. They were also the most obvious and imposing products of Spanish artisanry. But artisanry on a smaller scale had just as profound an aggregate effect on the lives of colonizers and colonized. The speed with which Spanish craftsmen appeared in America and went to work might at first seem surprising; but, given the opportunities for gain that the inevitable demand for familiar goods from *encomenderos* and other wealthy settlers offered, it is less so. Probably at least a tenth of Spanish settlers in Peru up to 1560 were active craftsmen, 800 or more of them.²⁷ In order of descending numbers, the following were present and at work: first, tailors and shoemakers; ironsmiths; builders; then lesser numbers of silversmiths, barber-surgeons, muleteers, pharmacists, confectioners, arms makers, candlemakers, and musicians – and even a bookbinder.²⁸ Some specialization and subdivision of trades was initially lost in the transfer from Spain; but by mid-century there was little available there by way of craft products that could not be made locally, if a little more roughly, in Lima, Mexico City, or other provincial cities. This was quite clear in the Peruvian civil wars of the 1540s. Most of the muskets, armor, swords, horseshoes, and pikes used by all sides were locally made. Builders and carpenters soon imposed a familiar and enduring appearance on towns. "By 1545 the outward aspect of Lima was already what it was to remain for centuries, long lines of bare adobe walls broken by splendid wooden doors."²⁹ Gonzalo Pizarro, during his rebel leadership of Peru in the 1540s, kept a band of players capable of performing Renaissance part music. Musicians might double as dance masters. In 1552 a resident organ maker built a substantial instrument for Lima cathedral.

Clearly most of the goods produced by these immigrant craftsmen went to Spanish settlers. Equally clearly, incoming artisans could and did draw on rich native skills in working stone and cloth, and in the Andean case, metal as well; and to that must be added the ironworking abilities of Africans. But still a powerful acculturating influence flowed through Spanish craftsmen in the opposite direction, for many of the apprentices trained by arriving Spanish artisans were Indians and blacks. Spaniards were impressed by the speed with which Indians imitated and put into practice the new techniques (in, for example, weaving with Spanish looms, and making European-style musical instruments).³⁰ It is easy to imagine the pleasure of a native stonemason who, after working for years with nothing better than a cutting tool made of some harder stone, or at best bronze, was suddenly presented with a steel chisel. Thus Spanish techniques, tastes, and standards were quickly diffused into the new, miscegenizing culture.

FARMING

Most craftsmen worked in towns. And colonial towns, with their imported architecture and layout, and the various representatives of state power that they

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contained, as well as the new techniques of immigrant artisans, were obvious concentrations of acculturative influence. But signs and forces of change flooded across the countryside, too, as Spanish farming practice expanded. If introduced micro-organisms made themselves felt all too soon in epidemics, far larger fauna were not far behind: hens, goats, sheep, pigs, donkeys, cattle, horses. These animals, in fact, established themselves in most parts of Spanish America before any definite farming began. In the Caribbean islands they ran wild from 1493 on, and multiplied into numbers that not only fed the island settlers but provided vigorous stock to occupy the mainland. The rangy Spanish pig was as telling an ally of the conquistador as the horse – indefatigable, omnivorous, adaptable to almost any climate and vegetation America had to offer, from tropical swamp to highland desert. The rough and tough cattle were equally invasive; though more selective in their feed than pigs, they were better able to range fast and wide in pursuit of the grasses and grains they sought. Horses did likewise. The result was that the natural pastures of the mainlands, notably in northern Mexico and the plains of the Río de la Plata, but in many smaller areas elsewhere, saw enormous herds of feral animals appear with astonishing speed. The grasslands, never grazed before, and free, like those earlier in the Caribbean, of natural foes of the new animals, whether microbes, insects, or large predators, offered conditions in which numbers could grow tenfold in three or four years.³¹ This bovine and equine paradise was fleeting, to be sure; damage from overgrazing was apparent in northern Mexico in the closing decades of the sixteenth century. But the numbers, if no longer explosively increasing, remained high. In 1586, two cattlemen in the Zacatecas region branded 75,000 young steers on the large estates they had by then created to supply the mines with meat and leather, and to send stock south for sale in central Mexico.³²

Some of that northern beef ended up in the diet of Indians in the center. Indeed, as cattle multiplied, and the price of beef fell, meat protein became widely available to the mass of the native people for the first time in their history. It was never cheaper, perhaps, than in the mid sixteenth century. That, too, was perhaps the time when milk, cream, and cheese began to assume the large part they now play in the Mexican diet, a part surprisingly larger than in the Spanish. Again, though Indians themselves rarely raised the large imported animals, they took quickly to keeping pigs, sheep, and chickens, so that beef was not the only flesh now available to them. In sum, then, for native Mexicans the arrival of European domestic creatures brought a distinct modification of diet, and some change in the manner of producing at least its protein content, from cultivation of beans to raising of small animals. The same goes, with regional variations, for American natives elsewhere.

The dietary gain (if gain it was) came, however, at a cost. Indians had never had any need to fence crops in New Spain, since there were no large grazing animals to eat them. That changed with the arrival of cattle. By the 1540s destruction of maize plots was a severe problem in central and southern Mexico. Viceroy Mendoza went so far as to ban cattle ranches (*estancias*) in the valleys around Oaxaca; and commented to Don Luis de Velasco, his successor,

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"The Spaniards cry that I have ruined them, and they are right ... But ... if cattle are allowed, the Indians will be destroyed."³³ The difficulty was not just that cattle trampled and ate the maize, but that aspiring Spanish stockraisers had a strong motive to push Indians off their communal lands to secure grazing. This was a conflict wherever in America Indians farmed land that could feed cattle. In New Spain, Mendoza and Velasco, with crown backing, sought to mitigate it by urging the use of the northern plains for stockraising. To that end, Velasco issued land grants in the north during the 1550s. There was by then in any case the added attraction of the growing market for meat and hides in the northern silver mines, besides which it suited all Spaniards if cattle in any way disrupted the nomadic Chichimeca of the *altiplano*, who clearly posed a potential threat to Spanish presence there. This was not the only case in which cattle were explorers and conquerors alongside men.

Sheep could pose similar problems, especially since, in contrast to cattle, their main grazing areas remained in central and southern Mexico. Following the Iberian pattern, sheep breeders soon began to practice seasonal transhumance. By the late 1570s flocks totaling 200,000 or so moved for the winter from the surroundings of Querétaro to Michoacán; there was a similar movement from Puebla to the Gulf coast. Villages along these sheep runs suffered crop damage; indeed, a heavy concentration of sheep in a particular area could produce wider and more lasting damage in the form of soil erosion and loss of fertility.³⁴ Goats, too, with their predilection for eating plants right down to their roots, were a mixed blessing. They could thrive on steep, dry land where no other useful animal would, yielding meat, milk, and hides, but often, again, at the cost of erosion.

In general, indeed, it is hard to strike the balance of the pros and cons for the land and native people of the Spanish American empire of the introduction of European livestock. The equation inevitably contains too many imponderables for a clear answer to be evident. The gain of a new source of protein was offset by loss of crops and probably of some farming land (though some of the land occupied by grazing animals had clearly fallen largely or wholly vacant with the collapse of native numbers). Oxen, horses, donkeys, and then mules, complemented the sole American beast of burden, the llama of the central Andes. Elsewhere, they relieved humans of the age-old task of carrying freight, but again at a cost in land. The same was true of their provision of useful materials beside food. Leather for clothing, shoes and general use became cheap and plentiful, where it had largely been unavailable before. Wool there had never been before, except in the natural range of the Andean camelids, from Chile to southern Ecuador. The existence of some native groups was changed radically by the arrival of the new creatures. The nomads of the South American pampas became horsemen hunting the wild cattle that multiplied endlessly on those plains.³⁵ The Navajo of New Mexico and Arizona turned from raiding their farming Pueblo neighbors for a living to tending the *churro* sheep and weaving its wool. In the more central areas of complex culture the changes were perhaps less profound, more a matter of adding what was useful in the new than of radically changing the old. Nonetheless, a world with the European

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Less dramatically were the new food-peaches, pears, apples; cane; rice; and the s them for the Spania gave the world by w they made their ma eat many of these ne as the colonists, be them. It must have much good land fo disappointment in t more than fine whe The first major area the nearby Atilxco late in the next dec lands of the Bajío [and modern agricu amazed by the high farming for wheat, north into the mai prime motor. In the near mining centers found. Other, smaller west. By the end of bread they wanted, of long trends in M the *tortilla de harina* in food the distincti prepared in the thir

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domestic animals present must have seemed quite distinct from the previous one without them; as distinct, perhaps, as the times in Europe before and after the arrival of mechanical means of transport.

Less dramatically obvious than introduced fauna, but equally pervasive, were the new food-yielding flora that the Spaniards imported: citrus fruits, peaches, pears, apples, grapes, melons, bananas; onions, olives, radishes; sugar cane; rice; and the standard European grains, with wheat pre-eminent among them for the Spaniards. These were not an equal exchange for what America gave the world by way of new foods, beginning with maize and potatoes. But they made their mark on the American landscape; and Indians, if they did not eat many of these new vegetable foods, assuredly were soon familiar with them as the colonists, beginning with the *encomenderos*, demanded that they grow them. It must have been a great relief to the Spanish to discover in Mexico so much good land for wheat. The Caribbean islands and Panama had been a disappointment in that respect, and there was little a Spaniard hankered after more than fine wheaten bread. Highland New Spain proved ideal for the crop. The first major area put to wheat, in the 1530s, was the vicinity of Puebla, and the nearby Atlixco valley. Then, as the northward mining movement started late in the next decade, wheat followed, and so, gradually, "the former wastelands of the Bajío [became] converted into the most important, prosperous, and modern agricultural area of New Spain."³⁶ Travelers from Europe were amazed by the high yields that wheat farmers extracted from the Bajío. Its farming for wheat, and maize also, was another aspect of that inclusion of the north into the mainstream of Mexican life for which silver-mining was the prime motor. In the more distant north, wheat- and maize-growing appeared near mining centers wherever suitable soil, and water for irrigation, could be found. Other, smaller, wheat areas developed in Michoacán and elsewhere in the west. By the end of the sixteenth century, colonists generally had all the white bread they wanted, and newcomers commented on its quality. More indicative of long trends in Mexican history, though, was the appearance in the north of the *tortilla de harina*, the tortilla made from wheat flour. Here was symbolized in food the distinctively *mestizo* culture of northern New Spain: the alien grain prepared in the thin, flat form of the traditional Mexican maize bread.

In South America, too, land for wheat was abundant, although what was to become the largest and most famous wheat area of all, the wet pampa of Argentina, was not planted with it until the nineteenth century, for lack of demand. On the western side of the continent that the Spanish mainly occupied, they were able to take advantage, like the native cultures before them, of the effect of varying altitude on local climate to create niches for wheat-growing. The coastal valleys of central Peru yielded enough of it, under irrigation, to supply the coastal towns in the sixteenth and seventeenth centuries. Cochabamba and other mid-altitude basins sent wheat to Potosí and other highland mining centers in Charcas. Central Chile, with its Mediterranean climate, proved ideal wheat territory, and exported large amounts of it to Peru in the eighteenth century. In the northern Andes, the elevated plains around Bogotá and Caracas gave the wheat that was locally needed.

Spaniards in America naturally began with a preference for familiar European foods, but where these were not available – and increasingly as the proportion of American-born settlers grew – they ate the New World's products as well. Maize became almost as basic for them as it was for natives; and in fact crossed the Atlantic to become a staple in north-western Spain in the seventeenth century. The potato was another American vegetable that Spaniards and colonists took to, after initial suspicion. Indeed it was the Spaniards who carried the potato from the Andes to Middle America, and also to Europe. Potato-growing began in the Spanish Basque country before 1600.³⁷ Other examples of American foods adopted more or less eagerly by colonials – chocolate, tomatoes, avocados, chile, and so on – are legion.

Nonetheless, it was the settlers' preference for their familiar European foods, rather than their consumption of American crops, that proved disruptive to Indians. Moreover, this disruption was magnified by their importing alien notions about the tenure of land on which crops and animals were raised. The concept of private landholding had scarcely appeared in native American cultures. Only in the Inca and Aztec polities, at the pinnacle of the nobility, was there any trace of it, and that was a late pre-conquest development. The prevailing concept of land was that it was communal property, whose use could be allocated to individuals or families as long as they worked it. The Spanish, by contrast, though expecting to find some land attached to towns for communal uses such as grazing and charcoal-making, thought of it first as something to be possessed individually and exclusively.

Legal possession came from the monarch, since, as in the Reconquest, all areas newly conquered were defined as *tierras de realengo*, or royal territory. The crown might itself grant the land away, or, as more commonly happened in America, for reasons of distance, delegate the granting to some colonial authority – conqueror, governor, viceroy, and often, in the early days, to town councils. Generally these locally made grants became permanent only when the crown confirmed them.³⁸ Many Spaniards in sixteenth-century America asked for and received land in this way, from urban plots for a house to great swathes of countryside for grazing. *Encomenderos* were among the earliest applicants, since their allocations of Indians did not include land. Far from all the land that Spaniards acquired in the 1500s, however, came to them in this state regularized way. Some was bought from Indians, generally from *caciques* wrongfully alienating communal territory for personal gain or to secure cash with which to pay their people's tribute. And some was simply occupied (as when, for example, cattle strayed on to unassigned land), or seized. Doubtless, *de facto* occupation became more tempting and simpler as land fell vacant through population loss. But how much of the native community land coming under Spanish control was already unused for that reason, and how much was torn from active cultivators, is impossible to say. The proportion certainly varied by region. On the coasts of Peru and New Spain, for instance, where population losses were particularly severe, a larger proportion of the land taken by Spaniards had fallen into disuse than in the highlands of either area.

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In the 1590s the crown began to accept payments to regularize deficient or absent titles, in a program of "composition" that went along with a new policy introduced under Philip II of auctioning off, rather than simply granting, new land titles.³⁹ The change reflects not only Philip's always desperate need of funds, but also a growing demand for land in the Indies after the mid-century. Spaniards' interest in agriculture had risen slowly from a slender start in the immediate post-conquest years. At that time it seemed likely that Indians, working their own land, would supply as tribute most of what the colonists wanted by way of food. Some *encomenderos*, it is true, could be found even then who wanted land to produce food for sale, and still more of them, perhaps, whose desire for land sprang from a mixture of commercial and psychological motives, since landowning was part of the lordly lifestyle they aspired to create for themselves. But still, it was only when a blend of demographic decline and rising Spanish numbers made the notion of a tribute-based existence for settlers increasingly improbable that landowning became a general ambition grounded in economic reality.⁴⁰ This realization, in the middle decades of the century, coincided, furthermore, with the rise of the silver-mining centers in the central Andes and in northern New Spain. It was not hard to see in that development good possibilities for profitable sales of farm products.

And so there appeared in the latter part of the century a multiplicity of private farms, of varying sizes, owned mainly by Spaniards and *criollos* (American-born whites), but some also by blacks, *castas* (mixed-blood people), and undoubtedly some Indian leaders. In New Spain, the term *labor* was generally used for a small farm producing wheat and other crops for a local market. In the Andes, *chácara* (Quechua for sown land) had the same sense. In both areas, *estancia* meant land held for grazing. On many of these farms the owner might have been found physically laboring, though, if he were white, he would certainly rather not have done so. By the end of the century, the size of the holdings varied enormously, from a few dozen hectares of good arable land in a valley floor up to many thousands of sparse grazing on some arid plain. There was much buying and selling of land, so that farms' size and boundaries shifted constantly. With time, some of the larger holdings began to take on the look of the classic Spanish American rural estate, the *hacienda de campo*. But that term, and the institution itself – a sizable piece of land with a large house, laborers' living quarters, barns, stables, workrooms, and so on at its center, often self-supplying in foodstuffs and basic craft goods, indeed a rounded and self-contained community – belongs to the seventeenth century and later. The first colonial century saw the bases of the rural *hacienda* laid down in smallholdings. And in fact smallholdings remained an important social and economic element of the country scene of the colonies until the end.

Even the smallest private farm was likely to use European iron and steel tools to prepare the land for sowing, rather than the wooden digging sticks that were the most advanced native tilling tools. Larger places might well have steel-bladed plows (over 12,000 plowshares were sent from Spain to Mexico alone in 1597)⁴¹ and oxen or mules to draw them, though clearly there were many steep fields in the Andes and elsewhere where they would not serve. Still,

the great grain lands opened up by the Spanish in the first century, particularly those of New Spain, were worked extensively, as in Europe, rather than in the intensive American fashion. Yields per hectare were undoubtedly lower than the amazing levels sometimes achieved by native methods (with maize around the lakes in the Valley of Mexico, for example); but the use of animal-drawn tools raised the yield per unit of human labor expended on cultivation. The outcome was that European-style farming made inroads into even the supply of traditionally native foods. By 1630, for example, Mexico City's maize came largely from colonists' farms around it rather than from Indian communities. This rise of commercial agriculture was promoted still more by the use of animal-drawn carts. It was simple and relatively cheap to move farm products to market over considerable distances – tens, even hundreds, of kilometers – where the terrain allowed carting roads to be made. If it had not been so, the existence of the mining towns on the Mexican *altiplano*, for example, would have been distinctly more tenuous than it was. But where carts could not go, mule trains generally could.

By 1600, then, many fertile regions of Spanish America, and some regions that were not so fertile, were well scattered with privately held smallholdings, farms, and nascent estates. Few of the owners were native people. These lands were worked by predominantly alien techniques to produce predominantly alien crops; though with time those methods were applied also to some native crops. Large, imported, animals were a key part of the new agricultural method, putting non-human power for the first time at the service of American agriculture. Large tracts of private land, particularly those unsuited by climate, terrain, or fertility to tillage, served as grazing for these exotic beasts – not merely those used for draught, but a far larger number raised to give meat and raw materials, particularly wool and hides. Meat in the amounts now available was an innovation in the American diet. Similarly, wool and leather became unprecedentedly common. The animals' numbers multiplied prodigiously in the new environment (almost 1.3 million cattle in New Spain by 1620, by one estimate, and 8 million sheep and goats combined).⁴² In among the private landholdings, to be sure, native communities survived, each with its communal lands, safeguarded by Spanish law, that its people still worked by pre-Hispanic methods. Native farmers grew crops for their own subsistence, and certainly also for cash sale, where marketing opportunities existed (as, for instance, in nearby colonial towns). But Indian towns and villages had lost at least some of their best land to Spanish purchase, theft, and incursion. They were also fewer than a century before, inevitably so as a result of demographic contraction, and of Spanish policies, enacted from the mid 1500s on, of gathering Indians into new "congregations."

NATIVE LABOR

One purpose of creating those congregations had been easier access to native labor. Indians were brought together so that they could then be more easily

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removed to work in tasks profitable to both colonists and the colonial state. This was one in a whole progression of schemes devised to exploit native labor. In fact, questions of how, and how much, Indians should work for Spaniards were among the most hotly debated issues of imperial governance in the sixteenth century. The questions were not merely the practical ones of how to use a rapidly shrinking resource, but also the moral ones of how to reconcile the required conversion, civilization, and good treatment of native Americans with the soon evident need to compel them to work if the colonies were to sustain their populations and produce profits.

On the mainland, *encomienda* was the first solution tried. *Encomenderos* were supposed to look to the spiritual and material wellbeing of the Indians granted to them in return for tribute received. That tribute, for several decades, might take the form of labor ("personal service" was the term used) as well as offerings in bullion or goods. The outcome, though, tended to be as in the Caribbean islands: excessive and uncontrollable demands from the *encomenderos*, and rare fulfillment of the obligation to enlighten and improve the Indians. That imbalance, among other causes, led to the crown's attack on the *encomienda* in the New Laws of 1542, and, seven years later, to a ban on personal service from Indians in *encomienda*. Though neither the *encomienda* nor labor exactions in it disappeared at this point, nor in some places for long afterwards, the 1549 order does mark the beginning of the end of *encomienda* as the prime source of native workers in the central colonial areas.

It was replaced in those regions by state-directed systems of draft labor, in which native communities were ordered to supply a small proportion of their grown men at fixed intervals for assignment to particular tasks. In the central Andes the mechanism was called *mita* (Quechua for "time" or "turn"), the term used by the Incas for their own system of obligatory state labor, and the Indians caught up in the Spanish scheme clearly saw in it a link with pre-conquest practice. Elsewhere the Spanish word *repartimiento* ("distribution") covered the draft system (one of several official uses to which that word was put in the colonies). In New Spain a parallel existed in the native institution of compulsory public work, *coatequitl*, which continued to operate in surviving Indian polities after the conquest, though the name was not applied to the Spanish draft system.

Examples of ad hoc drafts can be found in the mid-century. In 1550, for example, Viceroy Mendoza drafted Indians to work on Spanish wheat farms around the capital, to relieve a grain shortage. Again, after severe flooding in 1555 Viceroy Velasco put some 6,000 Indians to work in building dikes and diverting streams to protect Mexico City.⁴³ Similar cases occur in the Andes also in the 1550s and 1560s. But in both regions it was the 1570s that saw draft labor generally and uniformly set up, through the efforts of viceroys Toledo, in Peru, and Enríquez, in New Spain.

At first the drafts were not overwhelmingly onerous. In the Valley of Mexico, the "take" in the *repartimiento* created for farming was 2 percent of adult men during the growing season, and 1 percent in the rest of the year.⁴⁴ In the most notorious draft of all, the *mita* organized by Toledo for Potosí in the mid 1570s,